REVIEW ARTICLE

Guidelines for the laboratory diagnosis of syphilis in East European countries

Abstract

The present guidelines aim to provide comprehensive and precise information regarding the laboratory diagnosis of the sexually transmitted infection (STI) syphilis in East European countries. These recommendations contain important information for laboratory staff working with STIs and/or STI-related issues. Individual East European countries may be required to make minor national adjustments to these guidelines as a result of lack of accessibility to some reagents or equipment, or laws in a specific country.

Keywords

Eastern Europe, guidelines, laboratory diagnosis, syphilis

Introduction

The diagnosis of sexually transmitted infections (STI) in Eastern European countries is still suboptimal. After the fall of the Soviet Union, increased mobility, economic insecurity, poverty, and lack of political attention resulted in epidemics of syphilis and increased incidences of other STIs, including human immunodeficiency virus (HIV) infections. During the past decade, we have evaluated the performance quality of laboratory services, their correspondence to international evidence-based recommendations, and performed optimizations in STI-related projects in several Eastern European countries, such as Lithuania, Estonia, Russia, and Belarus. Many laboratories in Eastern Europe need to optimize, standardize, and especially implement the use of modern methodologies and quality management systems. However, the present international recommendations cannot be applied in Eastern Europe either because the recommended diagnostic assays are not available, the diagnostic strategies are too expensive, or due to country-specific legal aspects.

Syphilis remains a significant health problem in many countries of Eastern Europe. The laboratory methods used for the diagnosis of the disease in the region are diverse and, in many cases, have not been validated. As a result of this situation, we consulted with opinion leaders in the region and developed a consensus document, which, with minor modifications, could be adopted for use by national regulatory authorities.

Methods used to establish a definitive diagnosis

The generally accepted criteria for the laboratory diagnosis of syphilis are presented in Table 1.

Direct detection of the causative agent – *Treponema pallidum* – is an absolute criterion for the definitive diagnosis of syphilis.

T. pallidum may be detected in samples obtained from lesions or other clinical specimens using:

- **Dark-field microscopy**
- **Direct fluorescent antibody test for *T. pallidum* (DFA-TP)**
- **Validated nucleic acid amplification tests (NAAT)**
- **Other equivalent methods**

These methods allow for the direct visualization or detection of *T. pallidum* and confirm active infection, particularly in early stages when lesions are present.

Table 1. Laboratory criteria for the diagnosis of syphilis

Diagnosis	Methods used
Definitive	Demonstration of *T. pallidum* in clinical specimens by dark-field microscopy, DFA-TP, validated NAAT, or equivalent method.
Presumptive	Positive result by two types of serologic tests: • A positive non-treponemal test (e.g., RPR, MPR, VDRL) • A positive confirmatory treponemal test (e.g., TP-PA, EIA, CIA, or immunoblots)